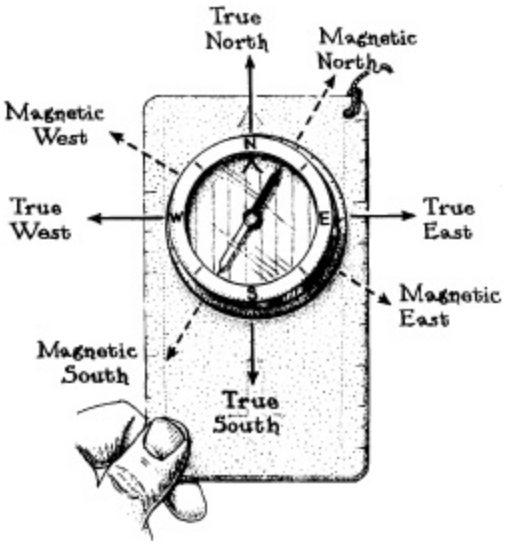


1.2 Magnetic north is a mixture of true north and true east, and true north is a mixture of magnetic north and magnetic west.

I, in the police station, were I not napping, would draw a rather different spacetime diagram to depict your car, your firecrackers, and the detonation (Figure 1.3c). I would plot the flow of time, as measured by me, vertically, and distance along Colorado Boulevard horizontally. As time passes, each firecracker moves down Colorado Boulevard with your car at high speed, and correspondingly, the firecracker's world line tilts rightward in the diagram: At the time of its detonation, the firecracker is farther to the right down Colorado Boulevard than at earlier times.

Now, the surprising conclusion of Einstein's logical argument (Box 1.1) is that the absoluteness of the speed of light requires the firecrackers *not* to detonate simultaneously as seen by me, even though they detonate simultaneously as seen by you. From my viewpoint the rearmost firecracker on your car detonates first, and the frontmost one detonates last. Correspondingly, the dotted line that we called "your space at moment of detonation" (Figure 1.3b) is tilted in my spacetime diagram (Figure 1.3c).

1.3 (a) Your sports car speeding down Colorado Boulevard with firecrackers attached to its roof. (b) Spacetime diagram depicting the firecrackers' motion and detonation from your viewpoint (riding in the car). (c) Spacetime diagram depicting the same firecracker motion and detonation from my viewpoint (at rest in the police station).

